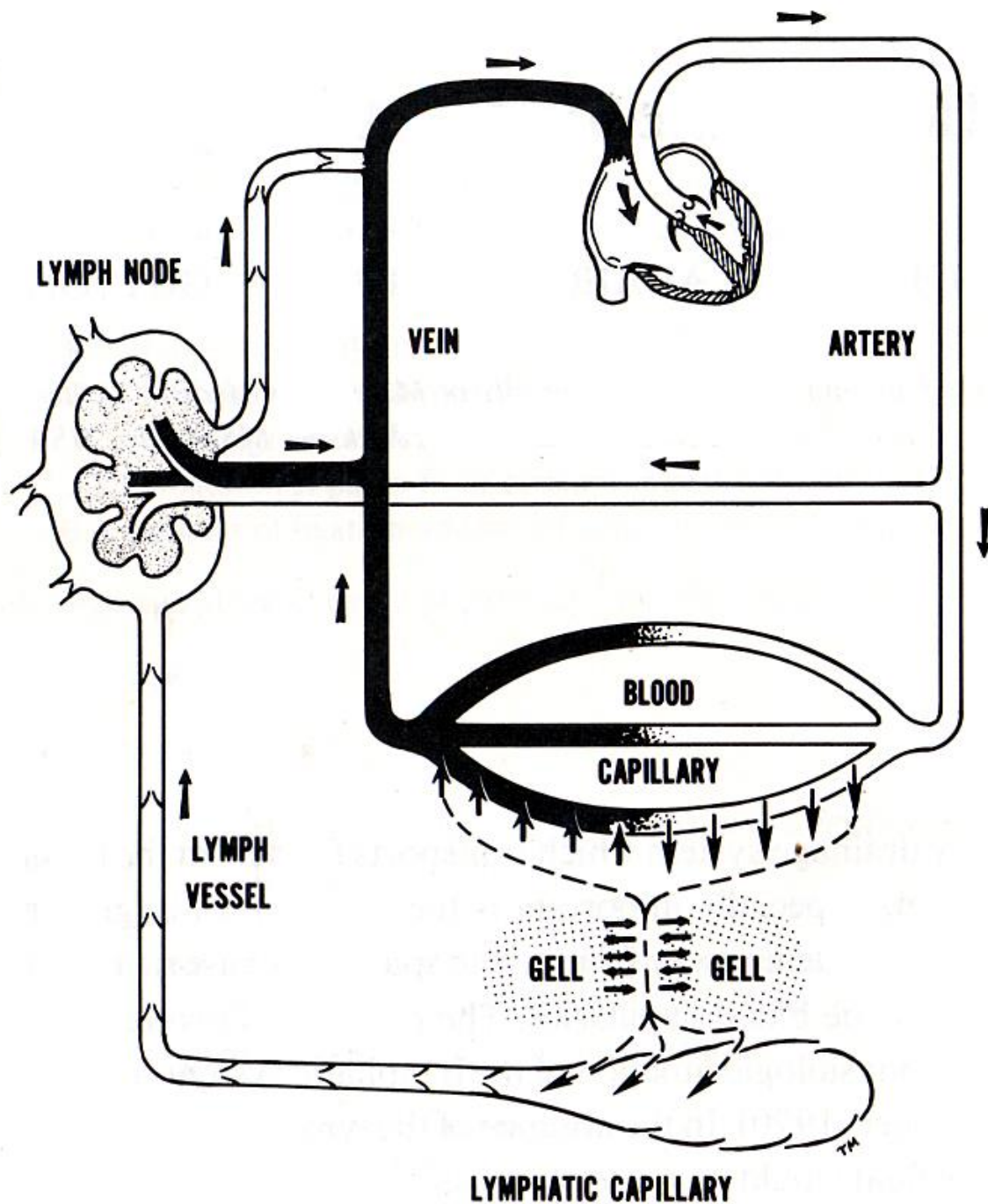


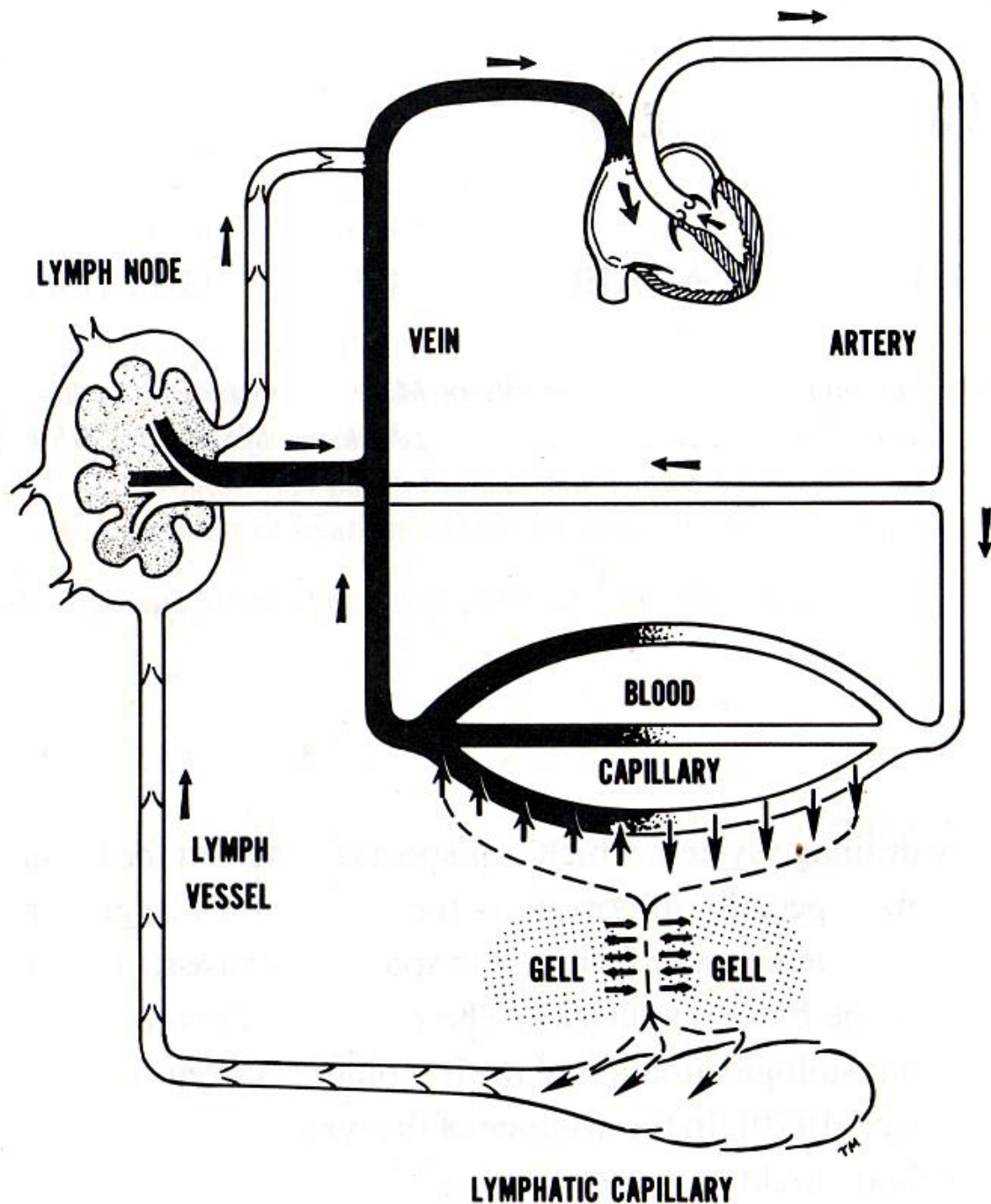
Session Objectives

- **Session Educational/Learning Objectives**
- 1. Understand the role of the Lymph system in the body's management of inflammation
- 2. Understand the consequence of Lymph dysfunction in acute and chronic disease
- 3. Learn the mechanics of lymph function, and the 3 separate aspects of Lymph drainage
- 4. Learn to recognize lymph congestion and areas of obstruction
- 5. Understand the physiology of the interstitium and the relation to the lymph system
- 6. Learn when to use the lymph pump



These (starling) forces do not account for the movement of fluid through the interstitial matrix (gel) where cellular respiration takes place

By the time fluid reaches the lymphatic, there is actually a small uphill gradient which must be overcome for lymph to form.



There is an interstitial mechanism, which includes tissue movement and fluctuation of extracellular fluid, that functions to promote interstitial interchange and facilitate cellular respiration

It operates by the principle of inertia

Lymph Regeneration

“Our recent work suggests that Lymphangiogenesis is a tissue response to interstitial fluid flow”

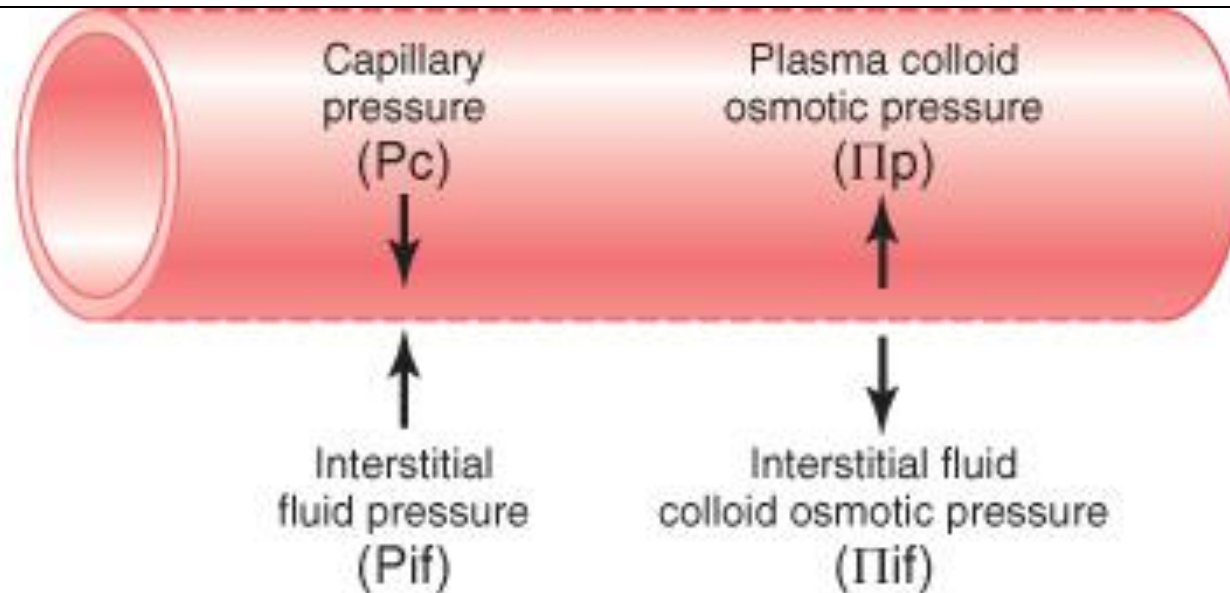
Swartz, M.A., et al

The Role of Interstitial Stress in Lymph Function and Lymphangiogenesis

Ann. N.Y. Acad. Sci. (2002) 979: 197-210

Boardman, K.C.

**Interstitial Flow as a Guide for Lymphangiogenesis
Circulation Research (2003) 92: 801-808**



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There is significant intravascular protein, and virtually no interstitial protein. This is a significant factor maintaining osmotic balance between the capillary and ECF.

The Lymphatics remove any protein that leaks into the interstitium, maintaining the osmotic balance of the ECF

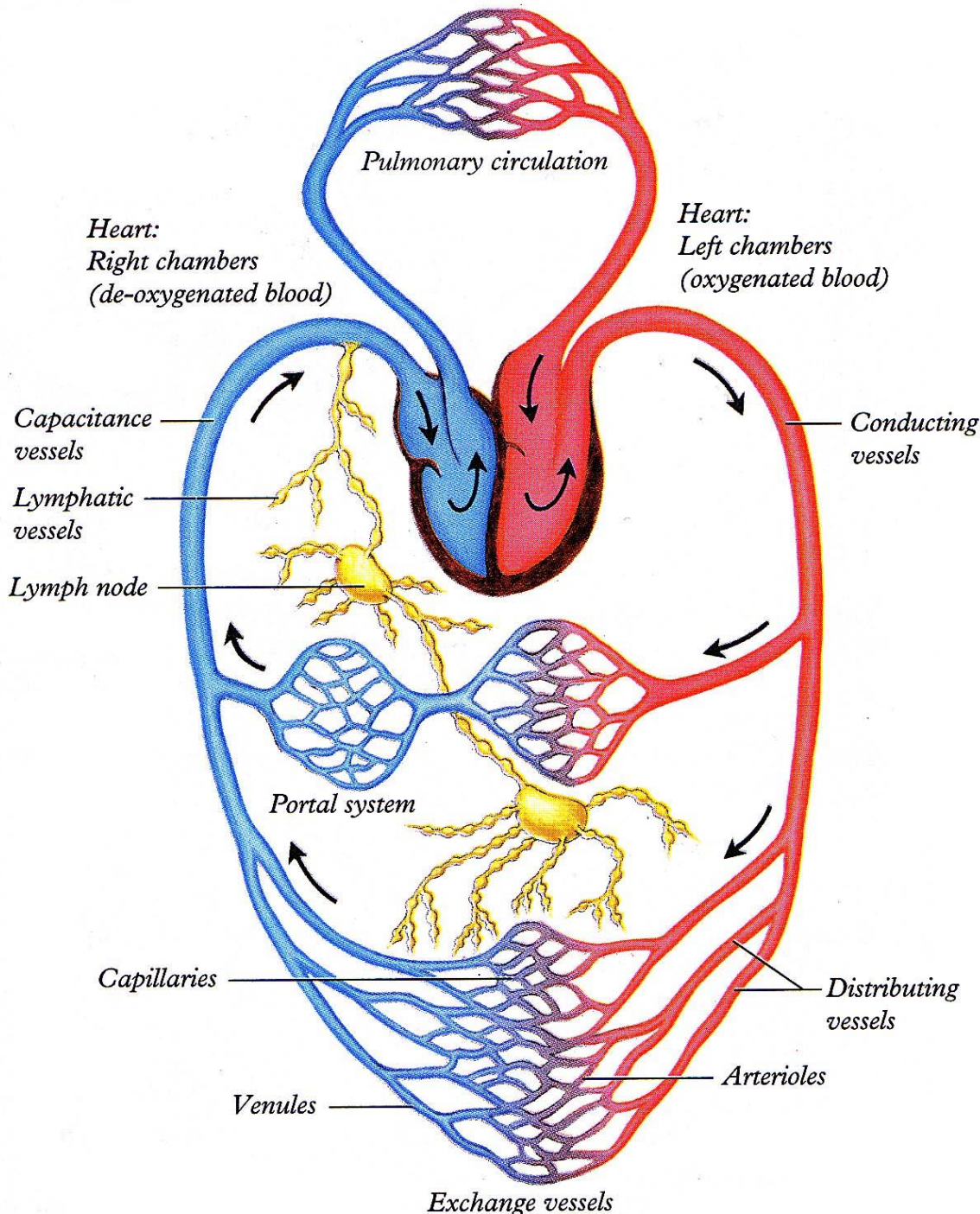
Role of lymphatics in inflammation

- Antigen – the lymphatics carry antigen to the lymph nodes where the immune response is initiated. This is particularly important in infections.

Role of lymphatics in inflammation

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- Circulatory Considerations

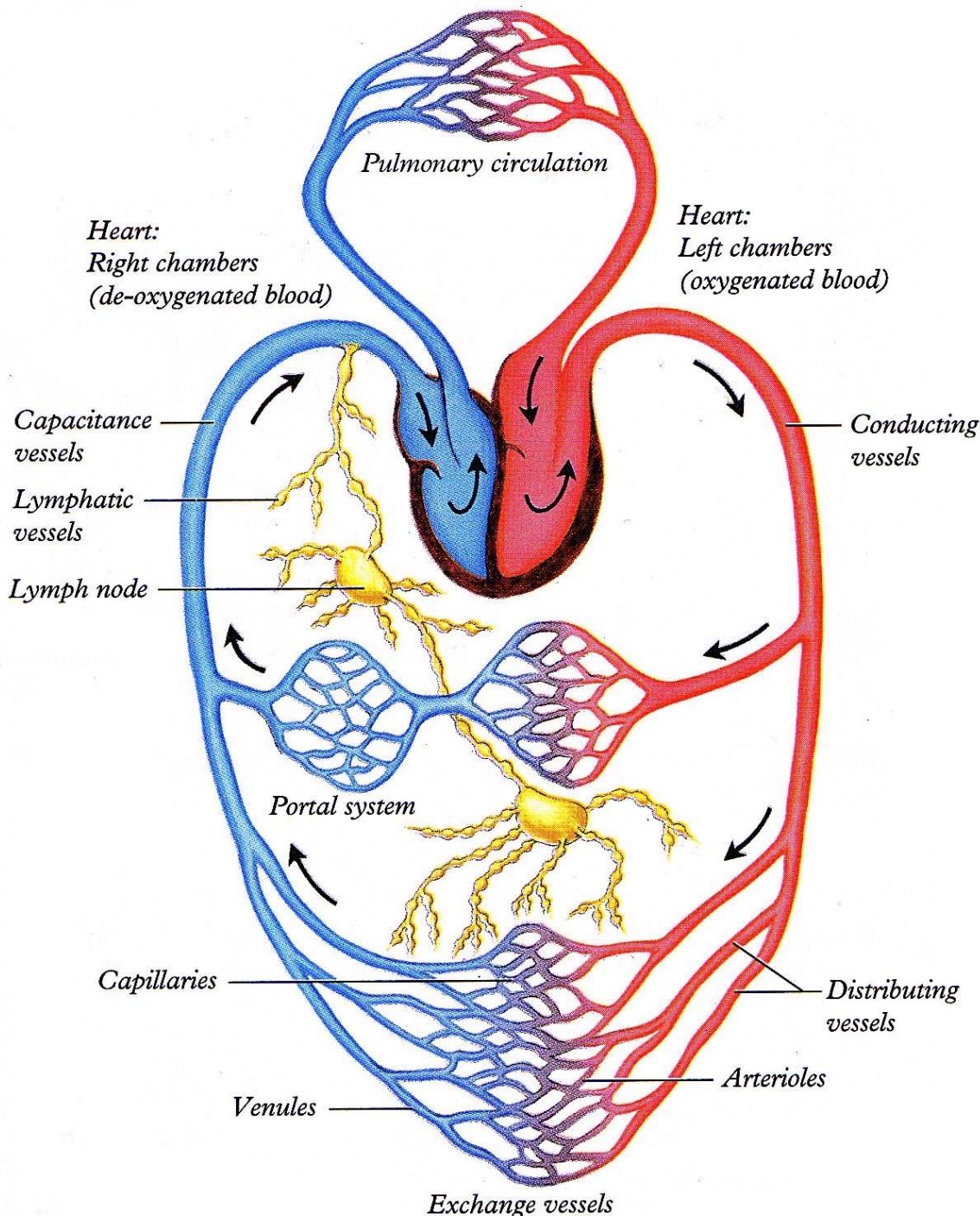
Circulation



- Inflammation produces a change in capillary permeability which allows a large protein efflux into the interstitium.
- This produces an inflammatory exudate, which can only be drained via the lymphatics.

Circulation

- Inflammation produces stasis of local capillaries, so that all circulation from an inflamed tissue is lymphatic.
- Once the tissue becomes saturated with fluid, circulation to the inflamed tissue will occur at the same rate as lymphatic drainage





- As an application, pneumonia is too much dirt in the wheels of the lungs. If so, we must wash it out; nowhere can we go to a better place for water than to the lymphatics.
 - A.T. Still Philosophy of Osteopathy, p 109

Role of lymphatics in inflammation

- Resolution of the Inflammatory process
- The lymphatics are central to the progression and resolution of inflammation
- Impaired or inadequate lymph drainage is implicated in chronic inflammatory diseases, as well as fibrotic diseases

Resolution of the inflammatory process

- Many if not most inflammatory mediators are broken down in the systemic circulation. They are brought there by the local lymphatic drainage.
- The progression and ultimate resolution of the inflammatory process with healing will depend on efficient lymphatic drainage.

Resolution of the inflammatory process

- Neutrophils are responsible for the lysozymes that produce most of the tissue damage during inflammation
- Their removal is critical to resolution of inflammation. They are removed by macrophages which phagocytise them
- These macrophages are removed from the tissue via lymphatic drainage.

Clinical correlations - Research

- What happens to normal tissue when lymphatics are removed
- Evaluation of lymphatics in diseased tissue
- Effects of disease in tissues when lymphatics are obstructed

Clinical correlations – Chronic Disease

- Lungs –
 - Interstitial fibrosis
- Heart –
 - Valve disease
 - Thickened fibrotic myocardium
- GI –
 - Inflammatory bowel disease
 - Pancreatitis
- CV
 - Atherosclerosis

Clinical Considerations

- Inflammatory Bowel Disease
 - Obstruction of Intestinal lymphatics produces bowel changes strikingly similar to IBD
 - Witte CL, Witte MH. 1984. Lymphatics in the pathophysiology of edema. In: Johnston MG, editor. Experimental biology of the lymphatic circulation. New York: Elsevier; p 167-88.
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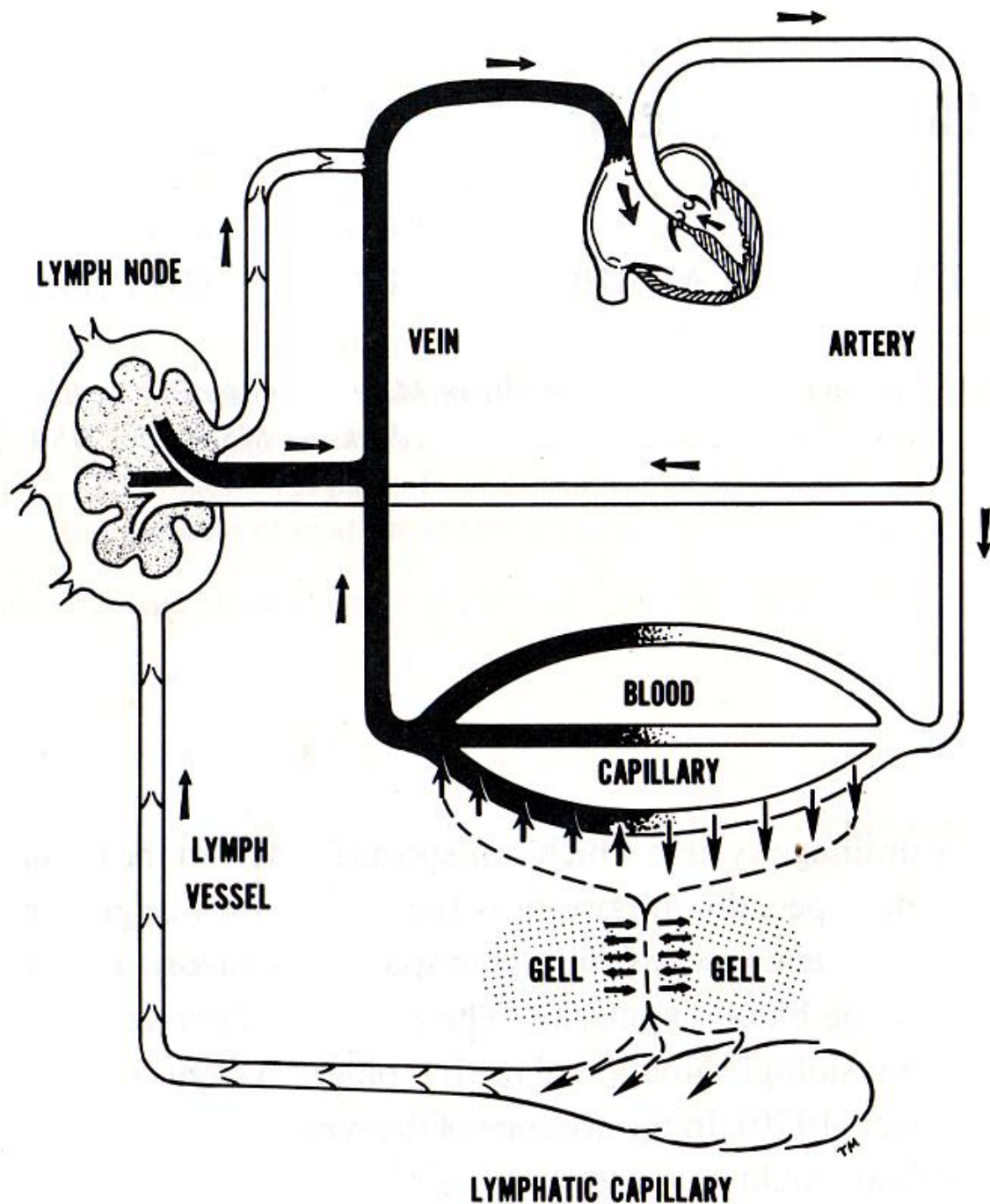
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- Inflammatory Bowel Disease
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 - The theory is that gut inflammation leads to fibrosis with consequent lymphatic obstruction, which leads to recurrent, more severe inflammation and subsequent bowel damage.

Clinical Considerations

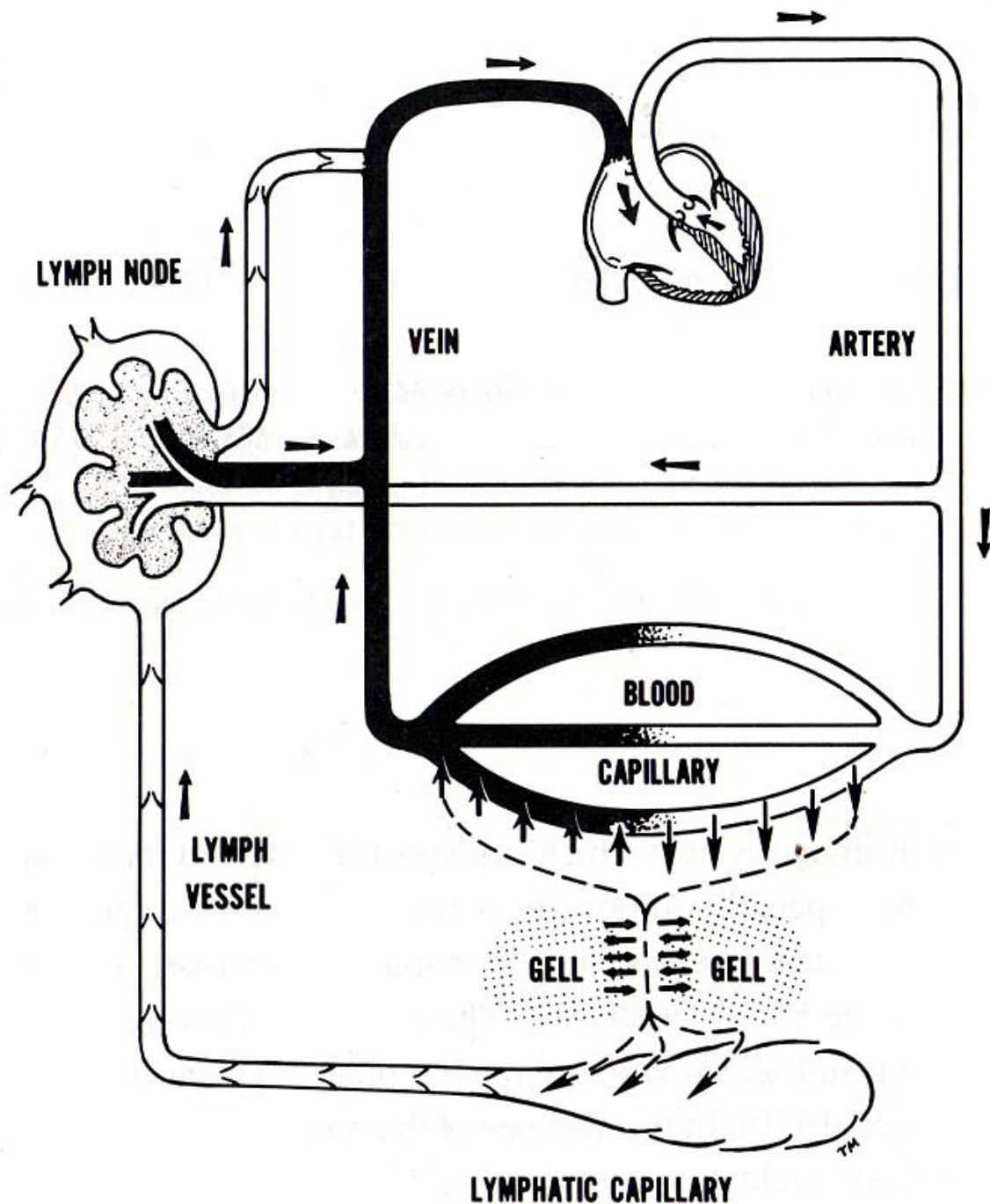
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There is an interstitial mechanism, which includes tissue movement and fluctuation of extracellular fluid, that functions to promote interstitial interchange and facilitate cellular respiration

It operates by the principle of inertia



Reduced fluid flux
(stasis) will reduce
lymph drainage and
increase fibrosis

Increased fluid flux
will improve lymph
drainage and increase
the regeneration of
lymph capillaries and
vessels.



- “If this be true we must keep them normal all the time or see confused nature in the form of disease, the list through. Thus we strike at the source of life and death when we go to the lymphatics.”
 - A.T. Still Philosophy of Osteopathy, p 108

Mechanics of the lymph circulation

- Formation – the movement of extracellular fluid into the initial lymphatic
- Vascular – the movement of lymph through the lymphatic vessels.
- Terminal – drainage of thoracic duct into the subclavian vein.

Mechanics of the lymph circulation

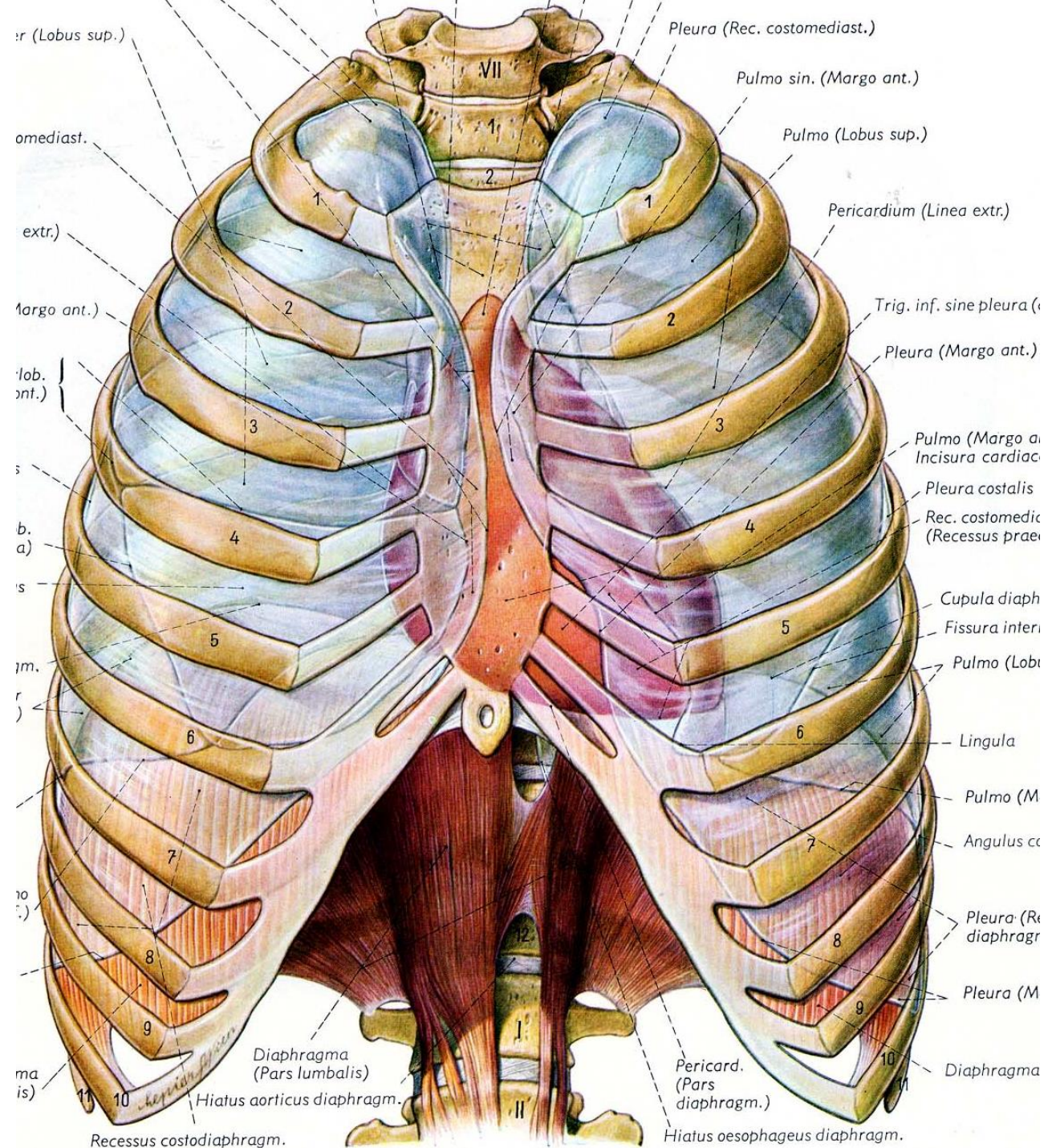
- Terminal – drainage of thoracic duct into the subclavian vein.
- The Respiratory/Circulatory model – J. Gordon Zink, DO

- “The treatment of the lymphatic system must always begin by securing ‘terminal drainage’ by way of insuring a good abdominothoracic venous pump. Venous stasis may precede lymph blockage, but it is the lymphatic disturbance that spells disaster to the tissues.”
- J. Gordon Zinc, DO The Osteopathic Holistic Approach to Homeostasis, AAO Yearbook, 1970, p. 4

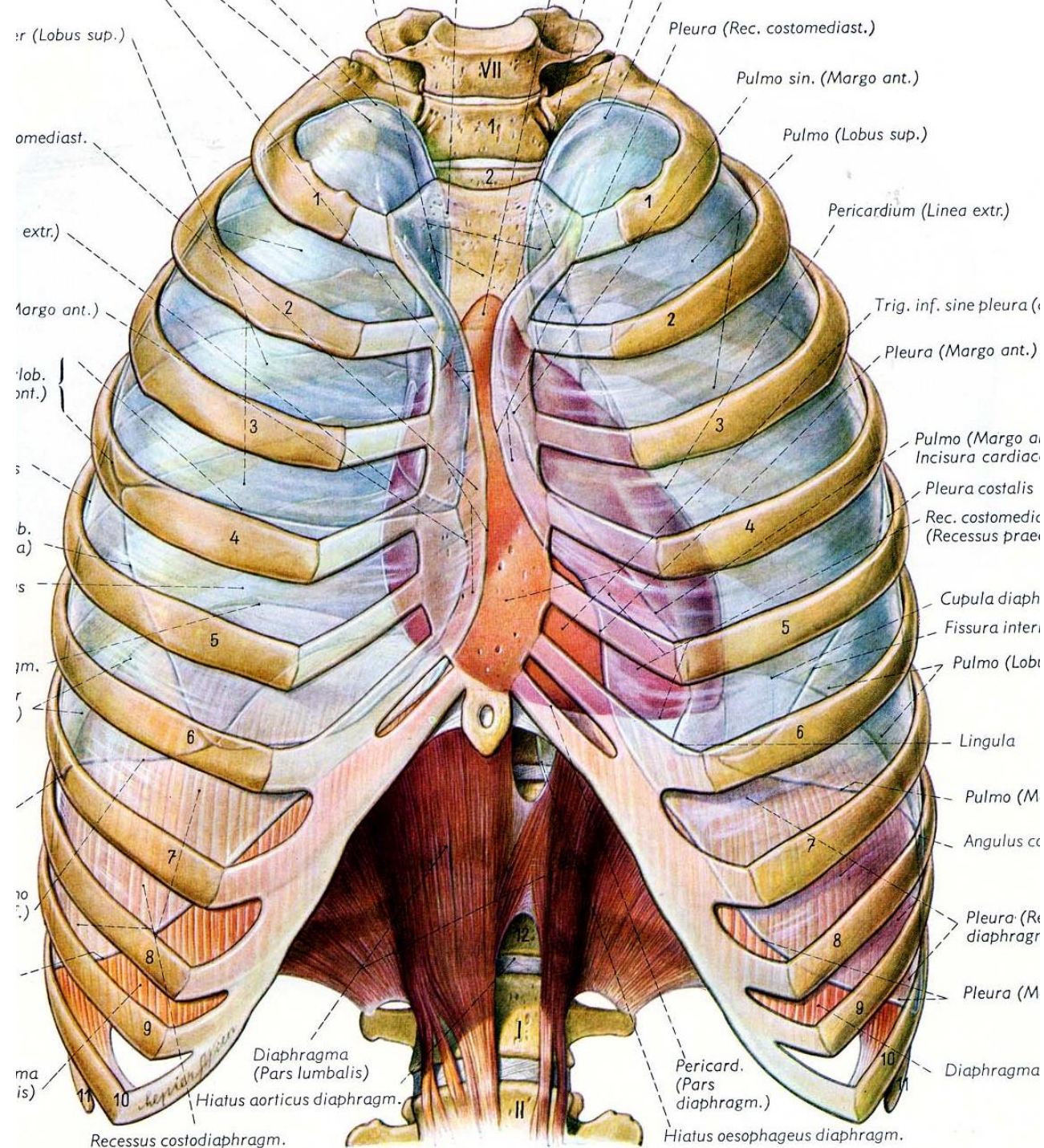
- “The cardiogenic aspect of circulation depends on the respirogenic aspect of circulation to complete the circuit. But that is not all; the most important feature is the fact that ‘terminal’ lymphatic drainage into the venous system is also dependent on this effective diaphragmatic respiration when the patient is resting.”
- J. Gordon Zinc, Respiratory and Circulatory care: The Conceptual Model, Osteopathic Annals, 1977 p. 11

Terminal lymph drainage

- Respiratory/circulatory function
 - J. Gordon Zink
 - 35 – 60% of thoracic duct flow is produced by respiration
 - Pressure changes in the thorax
 - Relation of crus to cysterna chyle
 - Venturi effect
 - Effect on lymph formation



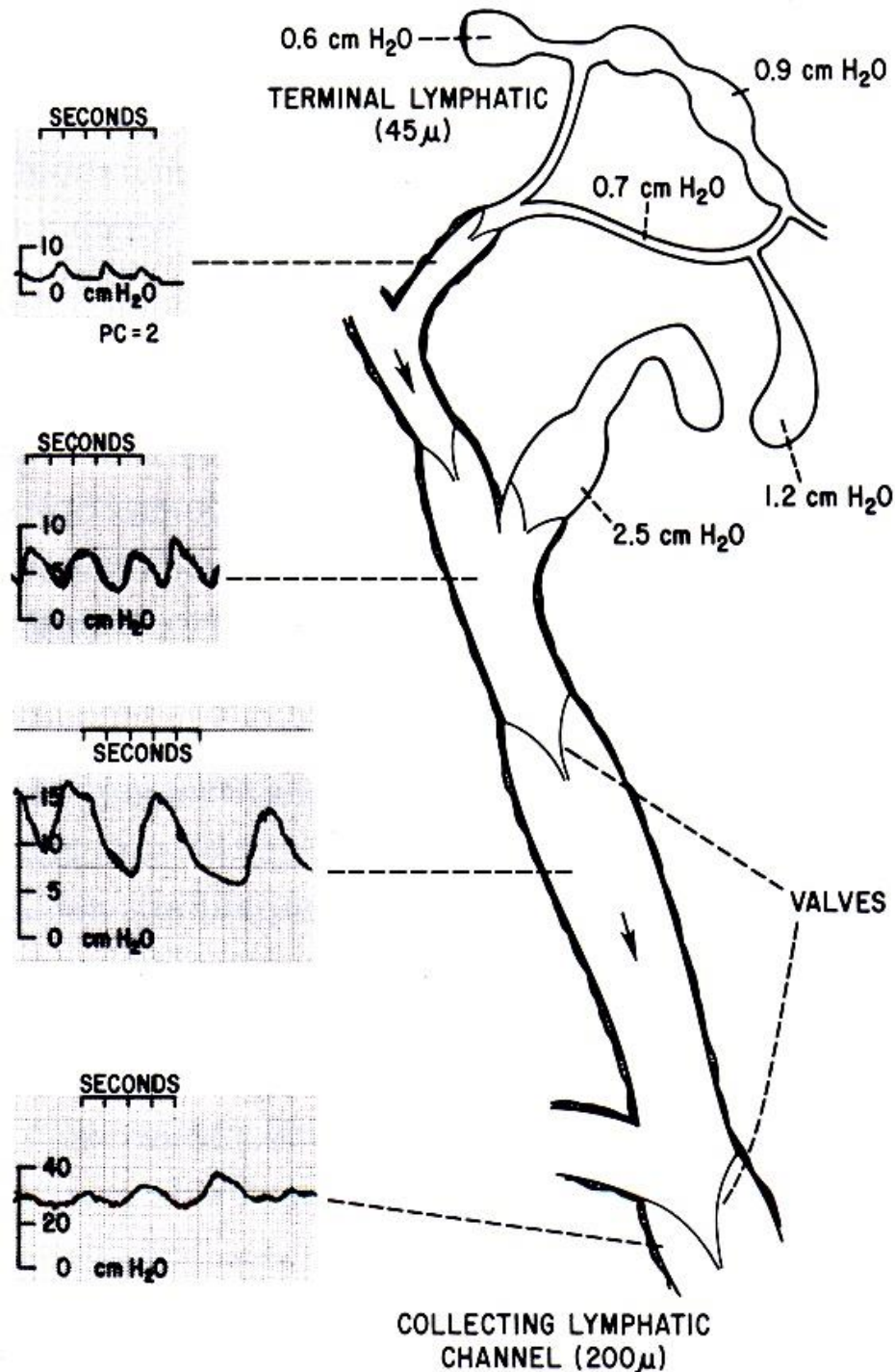
Negative
intrathoracic
pressures
produced by
respiratory
excursions
move venous
and lymphatic
fluid back to
the heart.



How effectively
is my patient
breathing

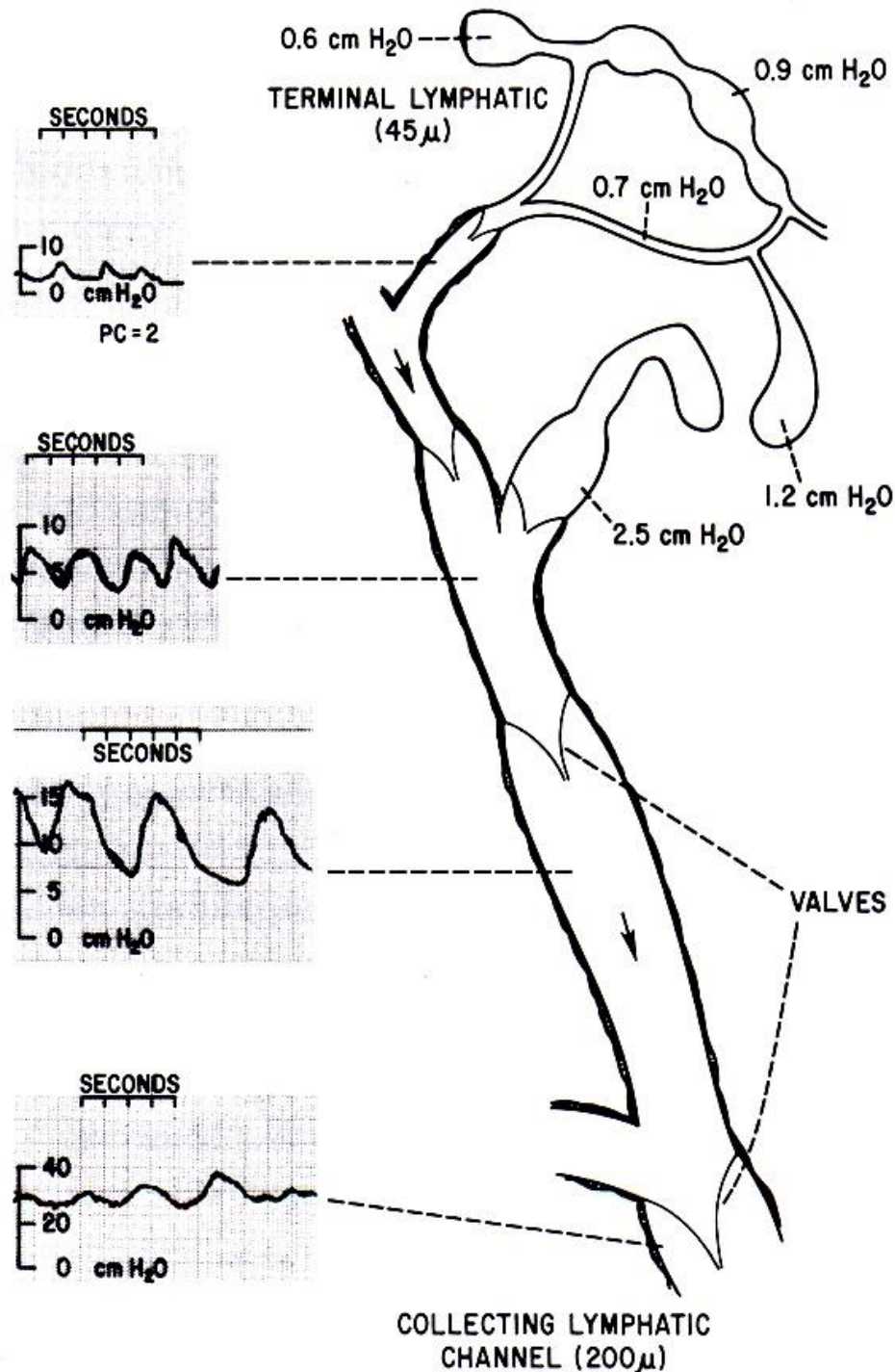
Vascular lymph flow

- Lymphangion – a unit with smooth muscle and valves
- Has intrinsic peristaltic action
- Can generate enough pressure to account for all intravascular lymph movement
- Can be obstructed with increased tissue tension/somatic dysfunction



Vascular lymph flow

- Eisenhoffer, et. al.
Effect of Outflow
pressure on lymphatic
pumping in vitro Am.
J. of Physiology 265,
1993
- Aukland & Reed.
Interstitial-lymphatic
Mechanisms in the
control of Extracellular
fluid volumes Physio.
Reviews 73, 1993



Vascular Lymph Flow

- Lymphangion – a unit with smooth muscle and valves.
- Has intrinsic peristaltic activity with pacemaker.
- Able to account for all movement of lymph through the vessels.
- Propulsion is dependent on formation

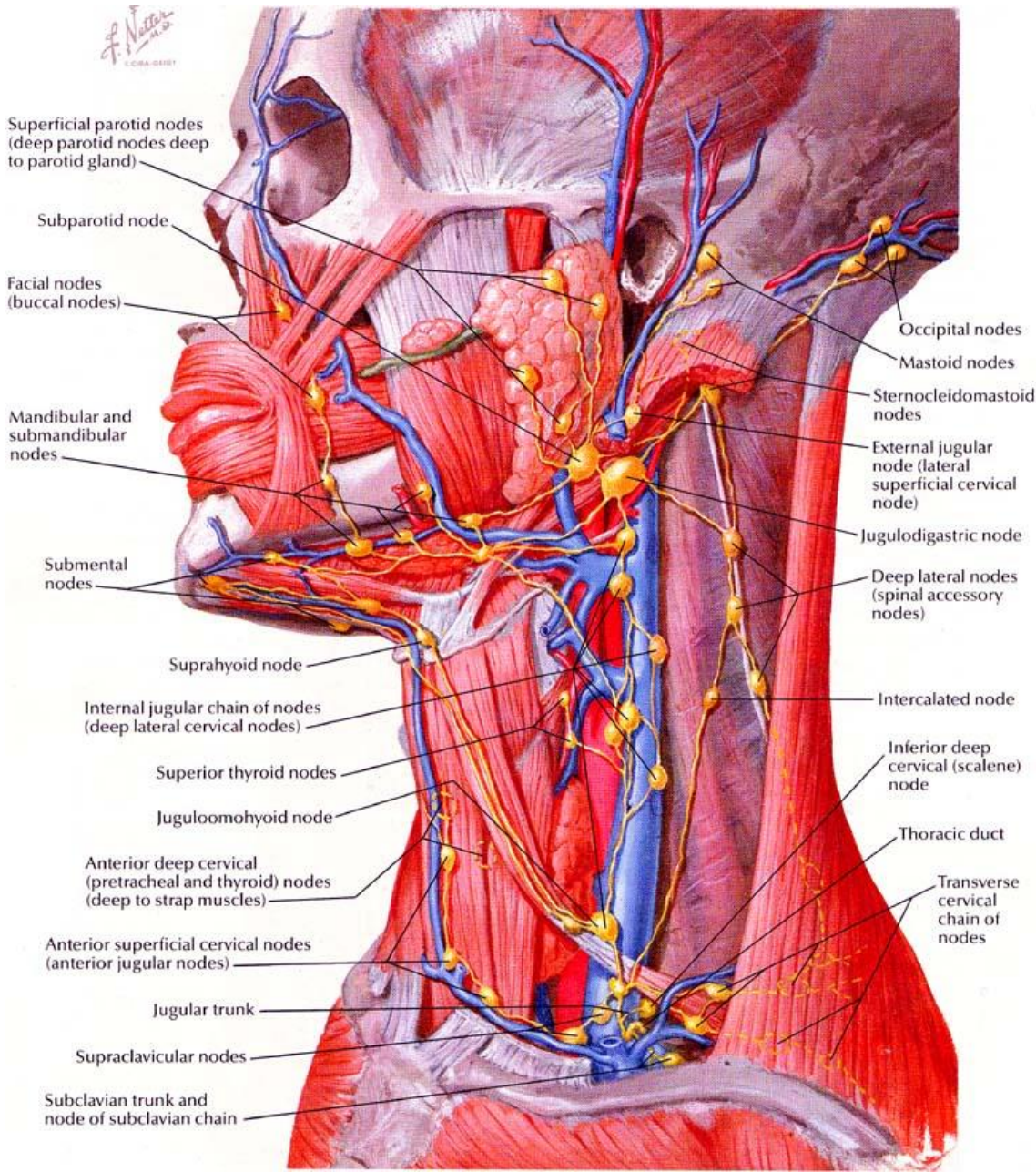
Vascular Lymph Flow

- The “path of drainage” should be evaluated and any dysfunctions which may be reducing lymph flow should be treated, before any “lymph pump” techniques are initiated.
- For each region, there will be “key areas” where the vessels are most vulnerable
- The most effective means of increasing lymph flow is to promote its formation

Areas vulnerable to obstruction

- “If one compares the various places where lymph nodes are found, then it can be established that they normally arise wherever the flow of lymph is obstructed at sites of kinking in the growing embryo. Such regions of lymph congestion arise wherever there are “bottlenecks” in the body; for example, in the region between the head and neck where the neck flexes, and at the creases where the limbs join the trunk in the vicinity of the armpit (axilla) and the groin (inguinal region).”
– Blechschmidt

Vascular lymph flow



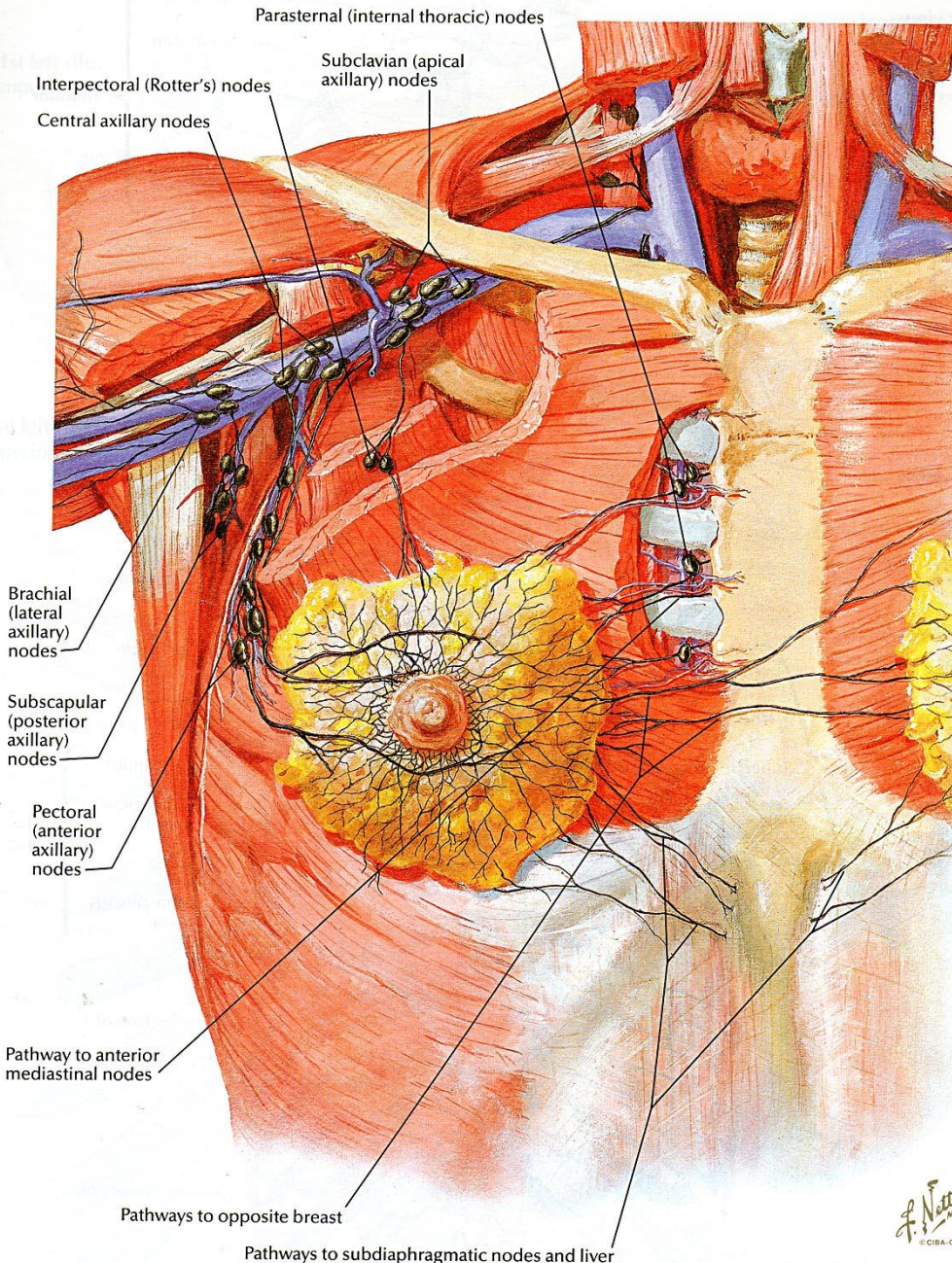
Can be obstructed with increased tissue tension/somatic dysfunction.

Most head and neck lymph must drain through the tight jugulodigastric area, between the mandible and the T.P. of C1

Vascular lymph flow

Can be obstructed with increased tissue tension/somatic dysfunction.

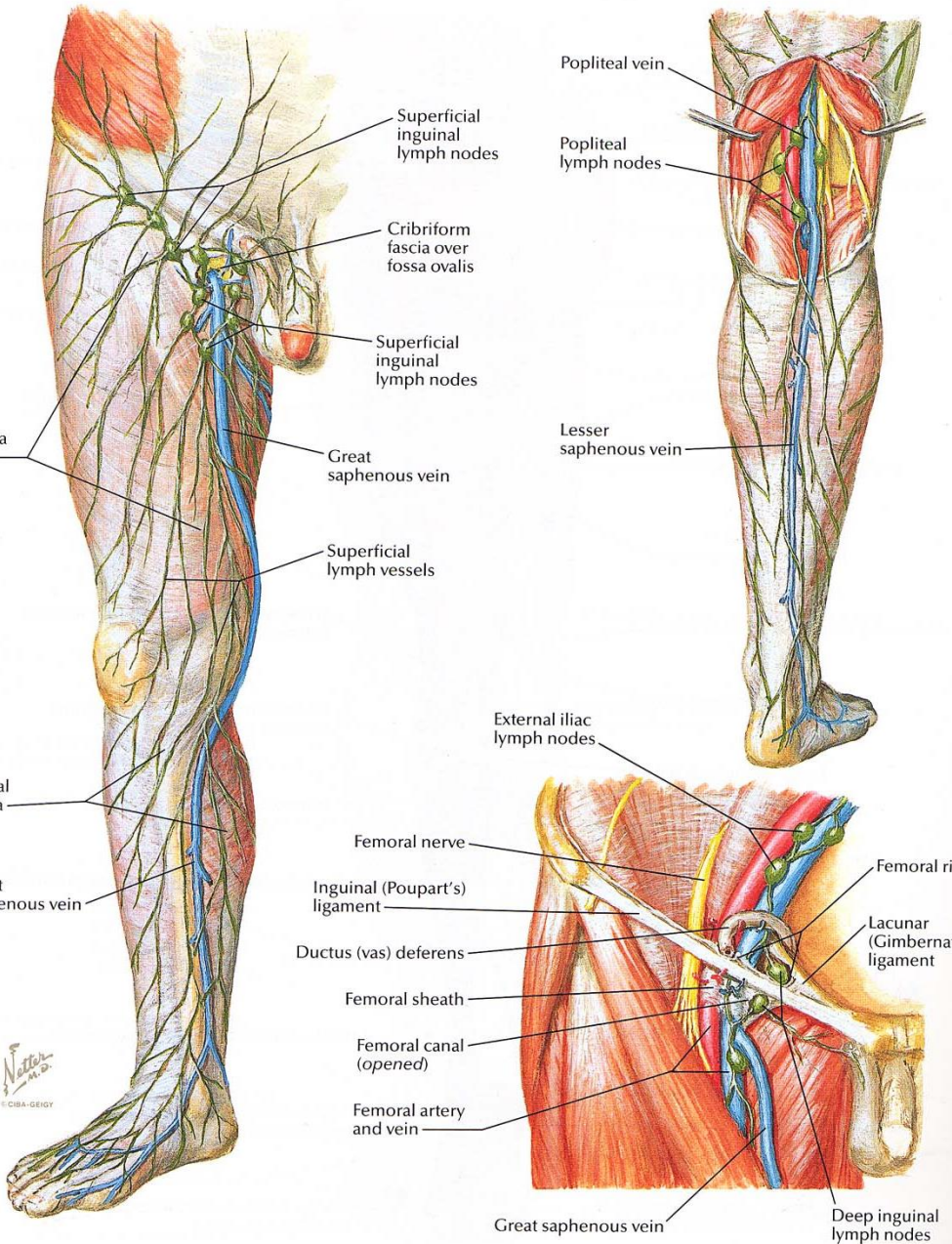
Upper extremity and breast drain below the axillary fold and then pass through the costoclavicular space.



Vascular lymph flow

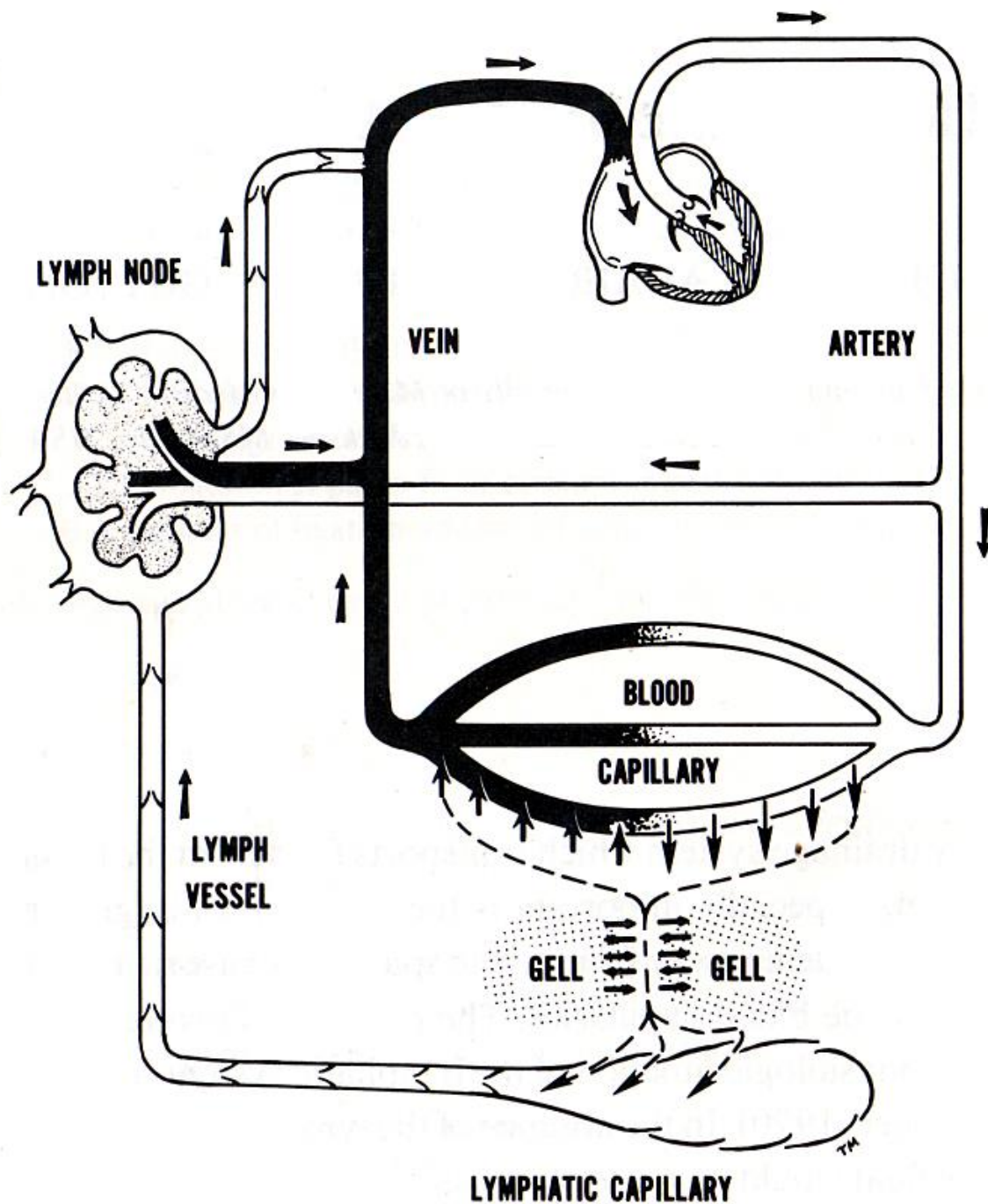
Can be obstructed with increased tissue tension/somatic dysfunction.

Lower extremity drainage is vulnerable as it passes through the femoral triangle, then the diaphragm



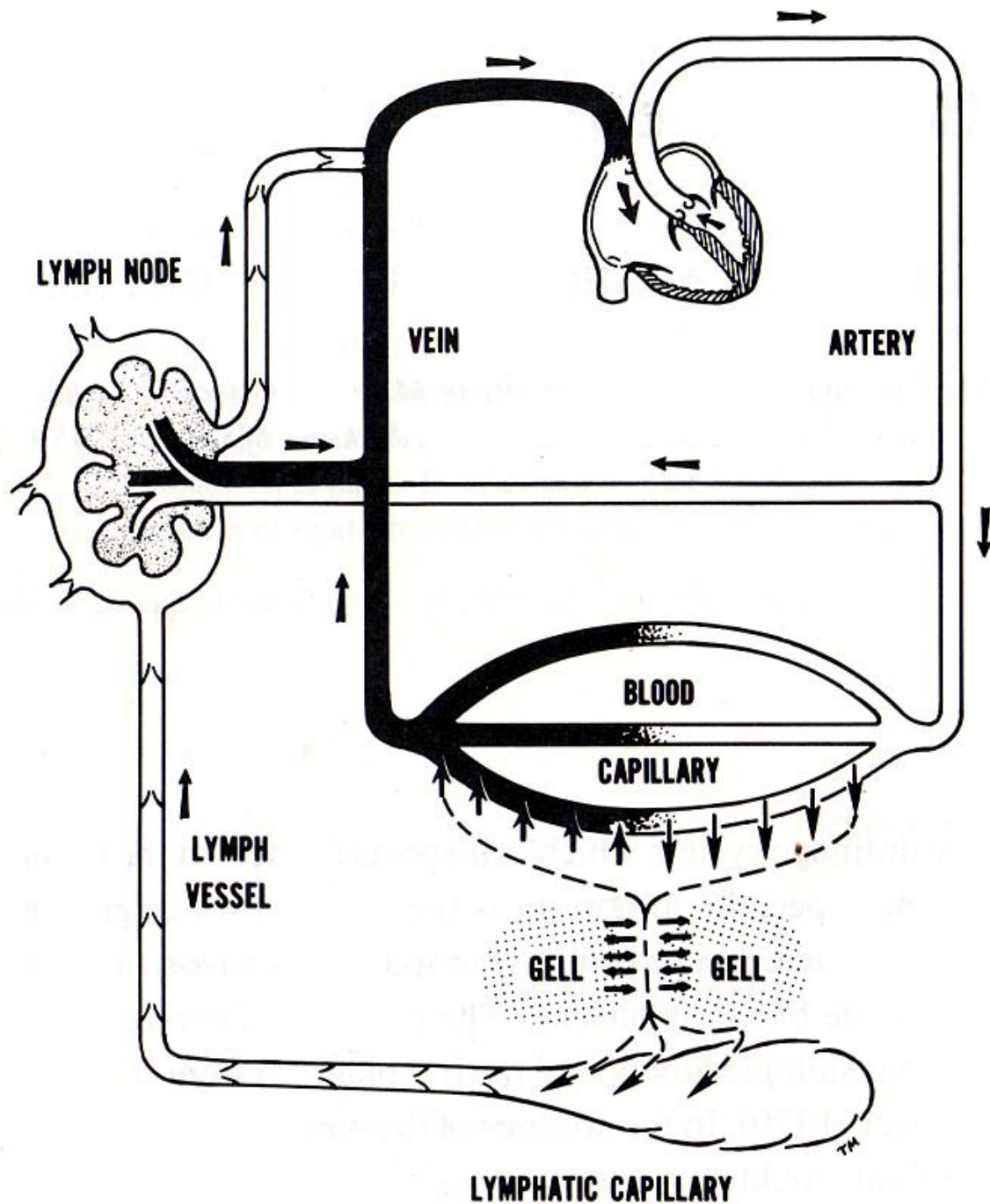
Lymph Formation

- There is a small uphill hydrostatic gradient which must be overcome for extracellular fluid to move into an initial lymphatic
- This may be the limiting factor in overall lymph flow
- External forces are necessary for lymph formation.



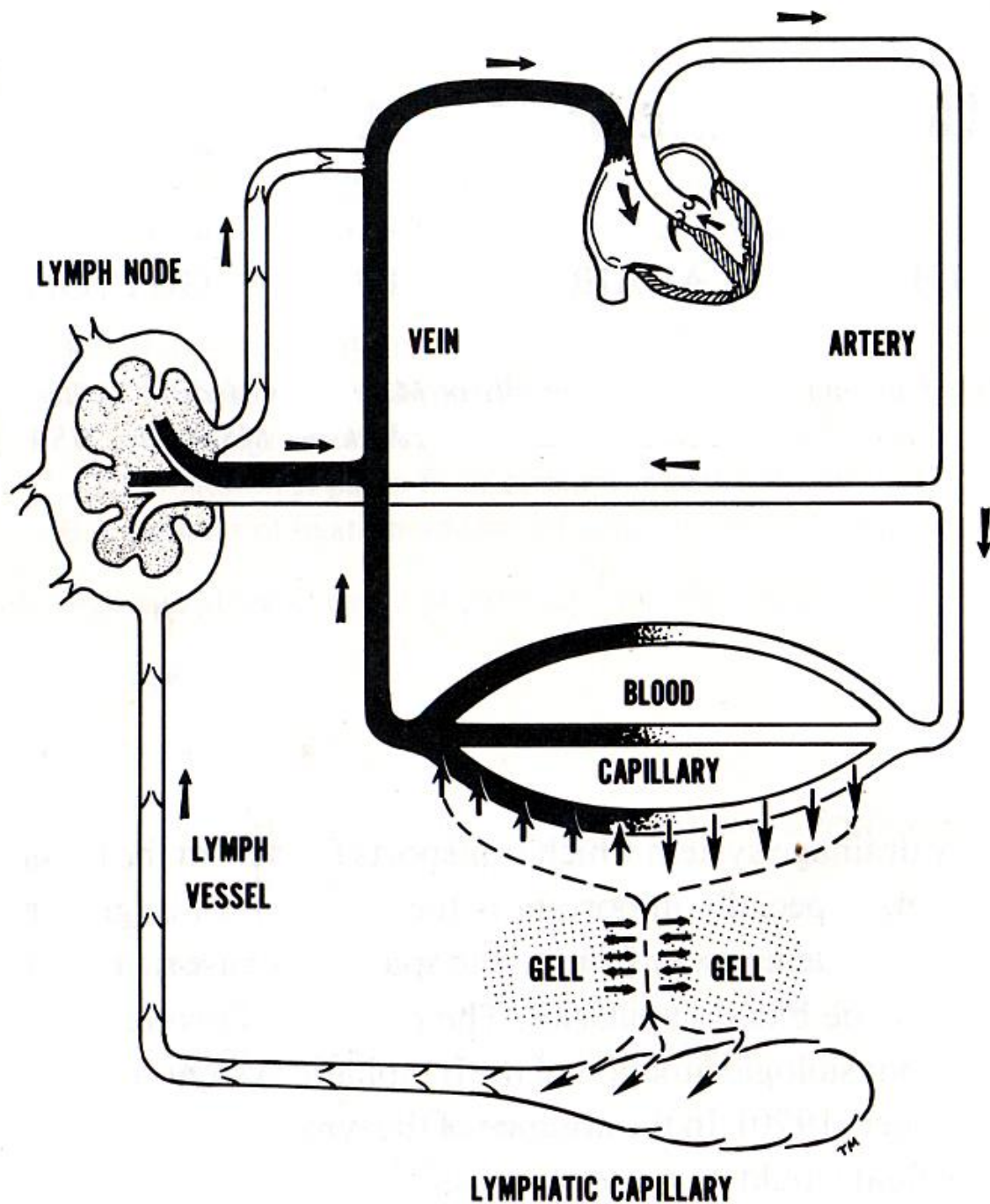
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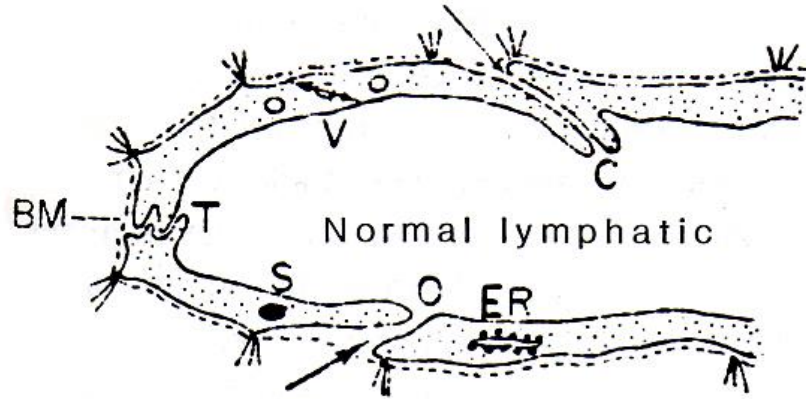


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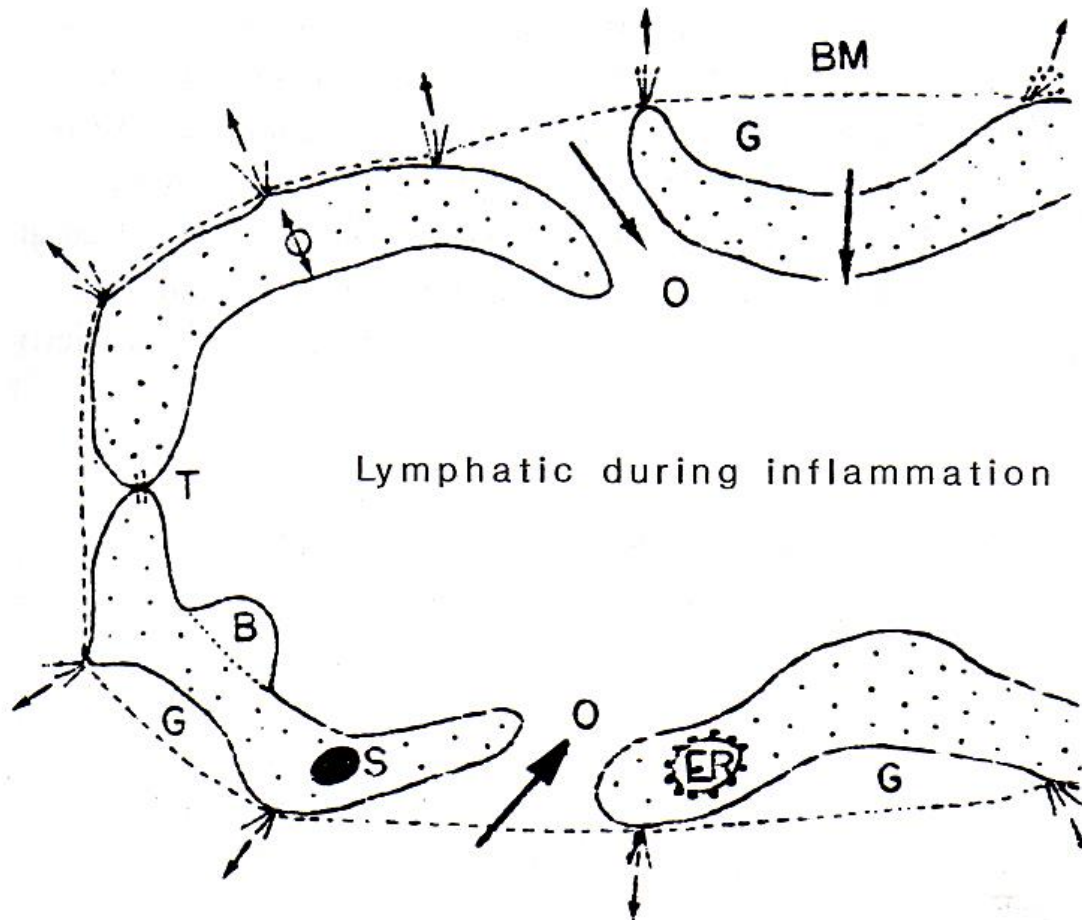


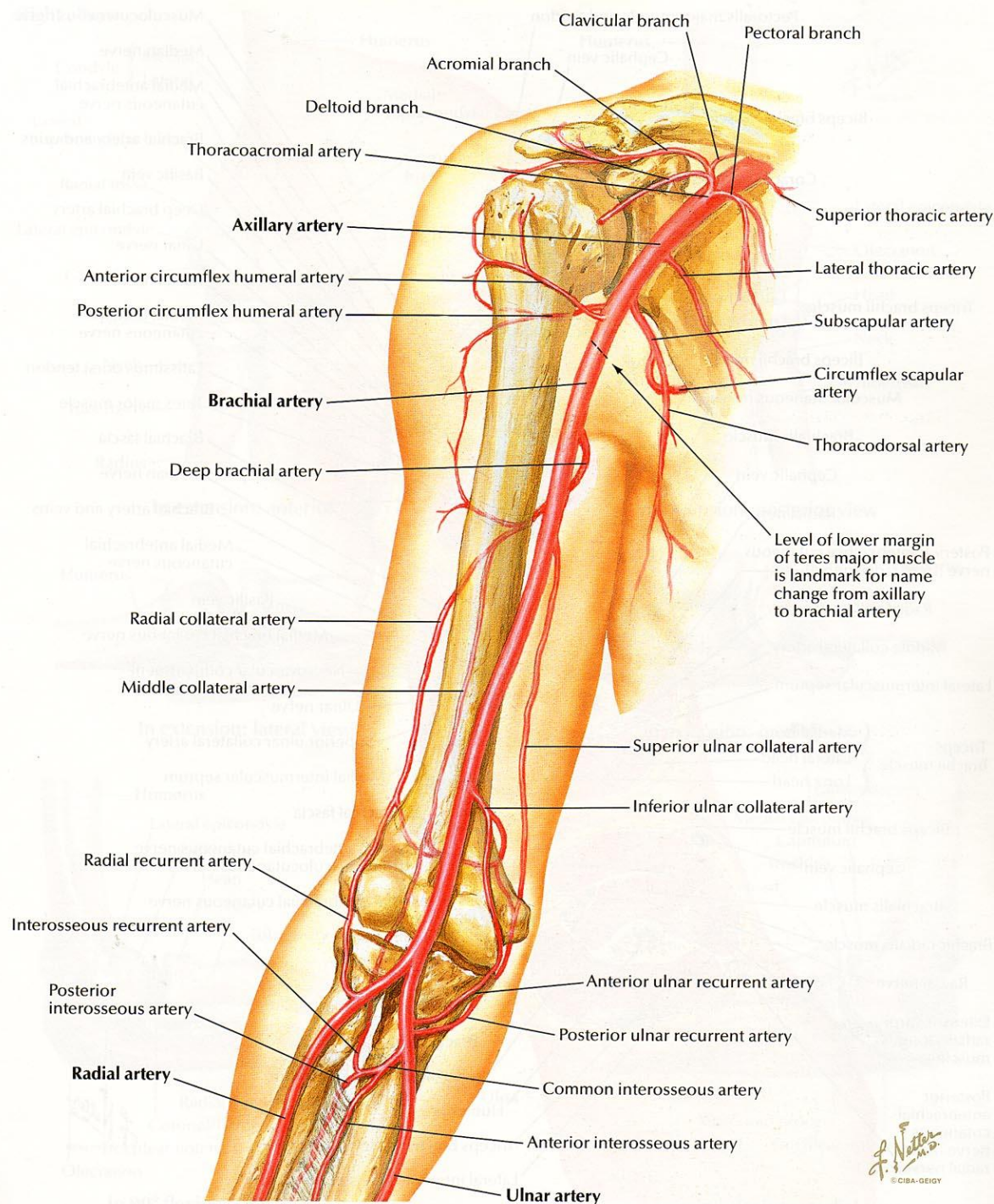
Lymphatics develop in the interstitial space, and in the presence of this interstitial mechanism, and develop and adapt to utilize these forces for the formation of lymph



Overlapping endothelial cells

Allow the ebb and flow of a fluid fluctuation to move fluid from the interstitium to the lymphatics





Arterial pulse creates a small fluctuation of fluid in the immediate vicinity of the arterioles. Many lymphatics are found in this area. Pulse has been shown to produce lymph in these vessels.

Lymph Formation

- There is a small uphill hydrostatic gradient which must be overcome for extracellular fluid to move into an initial lymphatic
- This may be the limiting factor in overall lymph flow
- External forces are necessary for lymph formation.
- This is where the “lymph pump” is applied

Lymph Pump

- A well-designed study demonstrated lymph pump to be vastly more effective increasing lymph flow when done over the area of swelling, ie. Lymph formation, compared to the area of vascular flow.
- McGeown, et.al. The role of External Compression and Movement in Lymph Propulsion in the Sheep Hindlimb J Phys (london) 387, 1987



- Without this cautious procedure, you' re patient had better save his life and money by passing you by as a failure, until you are by knowledge qualified to deal with the lymphatics.
 - A.T. Still
Philosophy of Osteopathy, p 105

Summary Slide

- Lymphatics clear proteins and large particulates from the interstitium to maintain osmotic balance
- This is critical in situations of inflammation, which includes most diseases
- Lymphatics not only drain the edema in inflammatory situations, but also the mediators, critical to the progression and resolution of the inflammatory process

Summary Slide

- Lymph drainage is a mechanical process with 3 distinct stages
- Terminal lymph drainage is the drainage of the thoracic duct and R lymphatic trunk into the central venous system. It is driven by respiration
- Vascular lymph propulsion has an intrinsic mechanism which includes smooth muscle in the wall of the vessel and valves. It can be obstructed by somatic dysfunction, especially in certain areas where the path becomes narrow
- Lymph formation is an active process requiring external force. Alternating movement and fluid fluctuation will fill the initial lymphatic, which will in turn stimulate the peristaltic contraction within the vessel to propulse it centrally
- This is where lymph pump is applied

Clinical Considerations

- Pancreatitis
 - Obliteration of Pancreatic lymph vessels can lead to fatal necrotic pancreatitis
 - Alexander, J.S. et.al. Gastrointestinal Lymphatics in Health and Disease. From Pathophysiology, 2009 Elsevier

Clinical Considerations

- Pancreatitis

- Lymph drainage has been shown to dramatically reduce the tissue damaging effects of pancreatic enzymes with obstruction of the main pancreatic duct in dogs.

- Witte CL, Witte MH. 1984. Lymphatics in the pathophysiology of edema. In: Johnston MG, editor. Experimental biology of the lymphatic circulation. New York: Elsevier; p 167-88.

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Clinical correlations - Myocardium

- Acute cardiac lymph obstruction produces edema, hemorrhage, and signs of ischemia.
- There is also reduced myocardial contractility, left ventricular stroke volume and cardiac output, and increased ventricular wall stiffness.
- Chronic lymphatic obstruction produces fibrosis and fibroelastosis of the ventricular wall.

Clinical correlations – vascular disease

- A wide variety of different studies are suggesting an important role for the lymphatics in the development of atherosclerosis.
- Atherosclerosis is presently accepted to be an inflammatory, proliferative disease rather than a degenerative.
- Many of the mediators which initiate/propagate this process must be drained via lymphatics

Clinical Considerations

- Pancreatitis

- In chronic pancreatitis, there is evidence of fibrosis with obliteration of lymphatics, reducing the capacity of lymph drainage of the pancreas, potentially predisposing to and/or increasing the severity of future attacks

- O' Morchoe, Charles. 1997. Lymphatic System of the Pancreas. Microscopic Research and Technique. 37: 456-477

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